

Assignment 2 (HW2) Report

Basic Information

- Name: Richard (Yunchao) He
- Report document:
https://docs.google.com/document/d/1BDw7-F90utmMCSdS60hVMx3flb_S_Uj42n4bt69tbLY/edit?

Discussions (Screenshots)

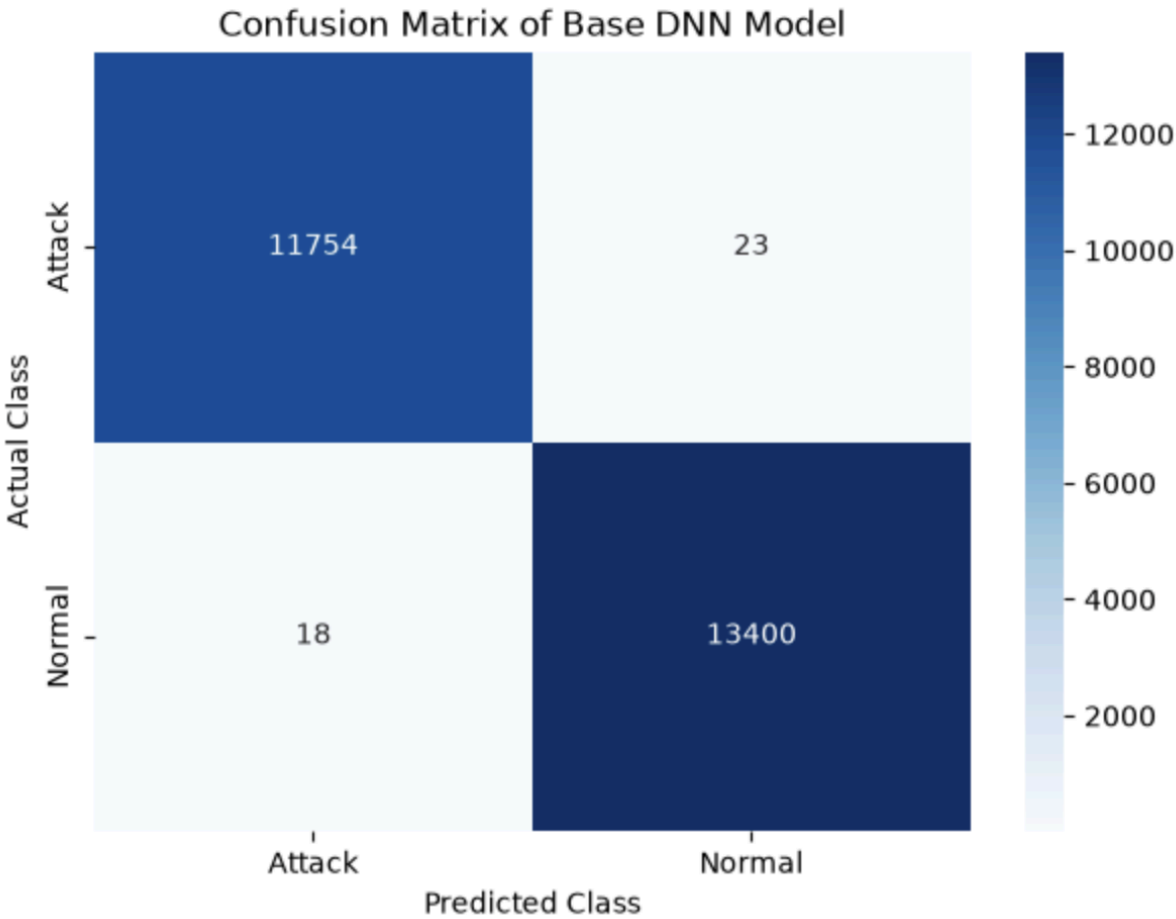
- Question 3(c): The screenshots of **base model** evaluation via **classification_report** and **confusion_matrix**

Classification Report:

	precision	recall	f1-score	support
Attack	0.9985	0.9980	0.9983	11777
Normal	0.9983	0.9987	0.9985	13418
accuracy			0.9984	25195
macro avg	0.9984	0.9984	0.9984	25195
weighted avg	0.9984	0.9984	0.9984	25195

Confusion Matrix:

```
[[11754  23]
 [  18 13400]]
```



- Question 4(b): The screenshots of **quantized model** evaluation via **classification_report** and **confusion_matrix**

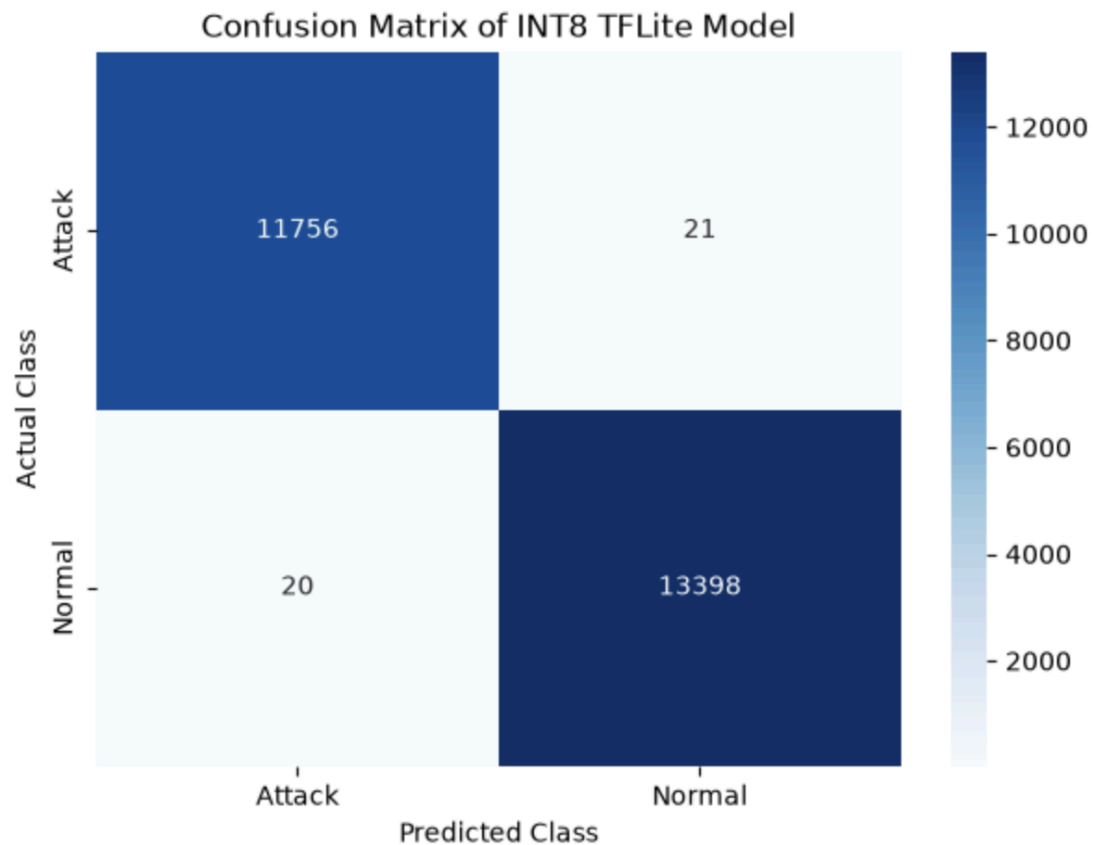
Classification Report for INT8 TFLite Model:

	precision	recall	f1-score	support
Attack	0.9983	0.9982	0.9983	11777
Normal	0.9984	0.9985	0.9985	13418
accuracy			0.9984	25195
macro avg	0.9984	0.9984	0.9984	25195
weighted avg	0.9984	0.9984	0.9984	25195

Confusion Matrix for INT8 TFLite Model:

```
[[11756   21]
 [   20 13398]]
```

INFO: Created TensorFlow Lite XNNPACK delegate for CPU.



- Question 5(c): screenshots of the printed Serial Monitor outputs for **five samples** with the original data in network_data.ino
 - Note: the result of five samples repeated again and again in Serial Monitor

The screenshot shows the Arduino IDE interface. The top bar indicates the project is named 'network_data' and the board is 'Arduino Nano 33 BLE'. The file explorer on the left shows 'network_data.ino' and 'network_model.h'. The main editor displays the contents of 'network_data.ino', which consists of a single line of code (line 131) containing a long list of 15 floating-point numbers. The Serial Monitor is open at the bottom, showing the output of the program. The output consists of two identical blocks of five lines each, each line representing a sample with its predicted and actual class values.

```

131 -0.01238515036740332, -0.02618002418454278, -0.018609896340735923,
132 -0.04122119759327531, -0.0028174939213690777, -0.0975309439715147,
133 0.15625477103722848, -0.35434285002584825, 1.602663889932865,
134 1.6051037177847889, -0.37436223991967527, -0.37443160310530493,
135 -1.4579091713193935, 0.03852930781964016, -0.37455970440553465,
136 -1.8058310079606816, -1.0266544017850858, 0.3313489400231479,
137 3.1073768183331816, 0.5877784745007281, -0.28910340026287856,
138 1.6087590765792643, 1.6189552037455606, -0.38763462350750655,
139 -0.3763870260680415, -0.6563667617603728, -0.11024922321249885,
140 -0.12470615670462065, 1.0873051789289023, -0.7362346401101137,
141 -0.00776224074056876, -0.004918644383724874, -0.08948642202040107,
142 -0.09507567152556495, -0.8092618187059747, -0.011663642603760032,
143 -0.036651869142258646, -0.024436507262009306, -0.01238515036740332,
144 -0.02618002418454278, -0.018609896340735923, -0.04122119759327531,
145 -0.0028174939213690777, -0.0975309439715147, 0.07765774737144172,

```

Serial Monitor Output:

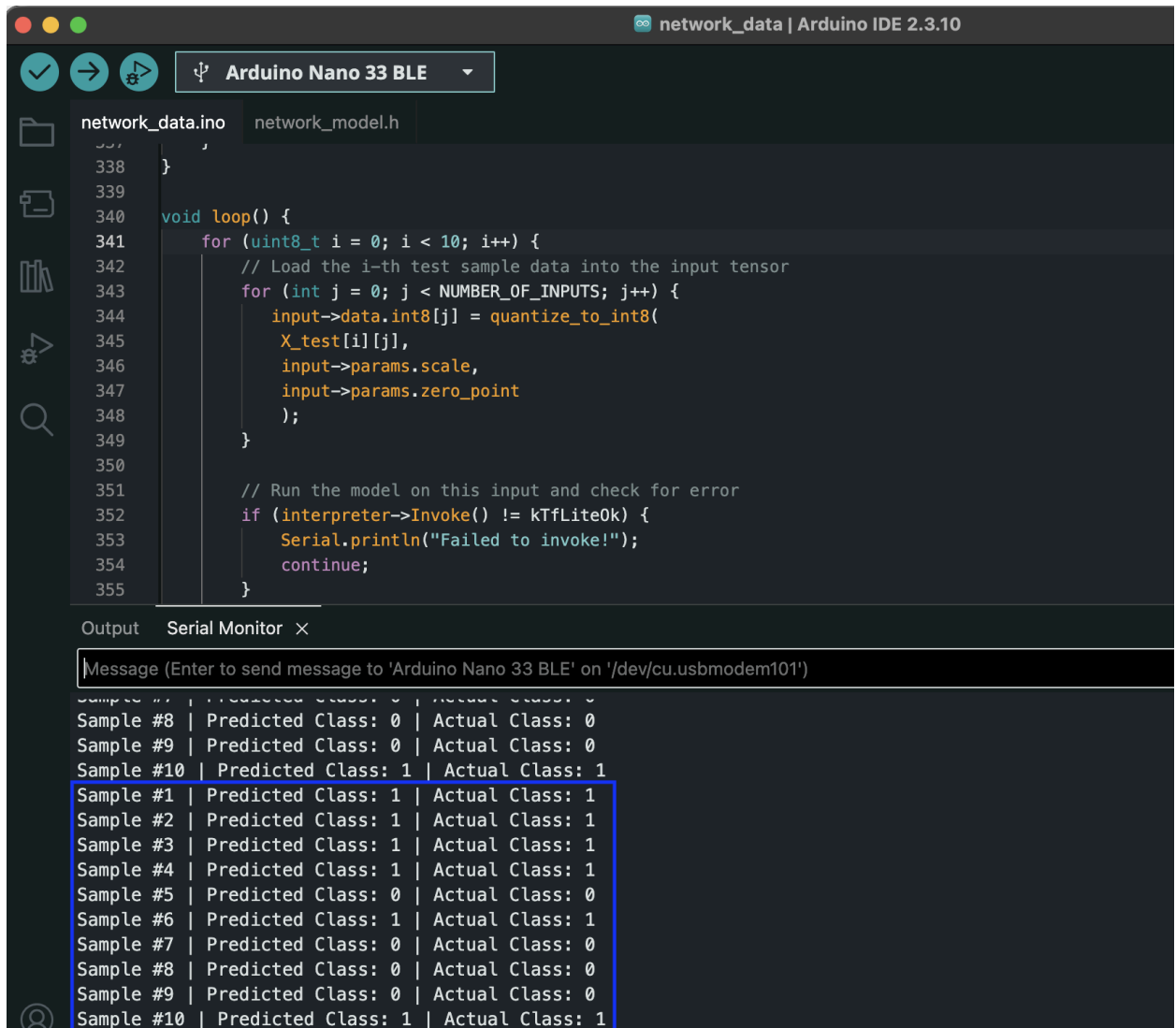
```

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem101')

Sample #1 | Predicted Class: 1 | Actual Class: 1
Sample #2 | Predicted Class: 1 | Actual Class: 1
Sample #3 | Predicted Class: 0 | Actual Class: 0
Sample #4 | Predicted Class: 0 | Actual Class: 0
Sample #5 | Predicted Class: 1 | Actual Class: 1
Sample #1 | Predicted Class: 1 | Actual Class: 1
Sample #2 | Predicted Class: 1 | Actual Class: 1
Sample #3 | Predicted Class: 0 | Actual Class: 0
Sample #4 | Predicted Class: 0 | Actual Class: 0
Sample #5 | Predicted Class: 1 | Actual Class: 1

```

- screenshots of the printed Serial Monitor outputs for these **ten samples** with the data generated from my jupyter notebook (and replaced the original data in network_data.ino)



The screenshot shows the Arduino IDE interface with the file `network_data.ino` open. The code defines a `loop()` function that iterates over 10 test samples. For each sample, it loads data into an input tensor, quantizes it, and runs a model to predict a class. The Serial Monitor displays the results for each sample, showing the predicted and actual classes. The output for samples 1 through 10 is highlighted in blue.

```
338 }
339
340 void loop() {
341   for (uint8_t i = 0; i < 10; i++) {
342     // Load the i-th test sample data into the input tensor
343     for (int j = 0; j < NUMBER_OF_INPUTS; j++) {
344       input->data.int8[j] = quantize_to_int8(
345         X_test[i][j],
346         input->params.scale,
347         input->params.zero_point
348       );
349     }
350
351     // Run the model on this input and check for error
352     if (interpreter->Invoke() != kTfLiteOk) {
353       Serial.println("Failed to invoke!");
354       continue;
355     }
356   }
357 }
```

Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem101')

Sample #	Predicted Class	Actual Class
Sample #1	1	1
Sample #2	1	1
Sample #3	1	1
Sample #4	1	1
Sample #5	0	0
Sample #6	1	1
Sample #7	0	0
Sample #8	0	0
Sample #9	0	0
Sample #10	1	1